

Problem

If $V_{ab} = 400$ V in a balanced Y-connected three-phase generator, find the phase voltages, assuming the phase sequence is:

(a) abc

(b) acb

Step-by-step solution

Step 1 of 2

(A) If $V_{ab} = 400$, Then,

$$V_{an} = \frac{400}{\sqrt{3}} \angle -30^\circ = 231 \angle -30^\circ \text{ V}$$

$$V_{bn} = 231 \angle -150^\circ \text{ V}$$

$$V_{cn} = 231 \angle -270^\circ \text{ V}$$

Step 2 of 2

(B) For the ACB sequence,

$$V_{ab} = V_{an} - V_{bn} = V_p \angle 0^\circ - V_p \angle 120^\circ$$

$$V_{ab} = V_p \left(1 + \frac{1}{2} - j \frac{\sqrt{3}}{2} \right) = V_p \sqrt{3} \angle -30^\circ$$

i.e.,

In the ACB sequence, V_{ab} lags V_{an} by 30°

Hence,

If

$V_{ab} = 400$, Then,

$$V_{an} = \frac{400}{\sqrt{3}} \angle 30^\circ = 231 \angle 30^\circ \text{ V}$$

$$V_{bn} = 231 \angle 150^\circ \text{ V}$$

$$V_{cn} = 231 \angle -90^\circ \text{ V} .$$